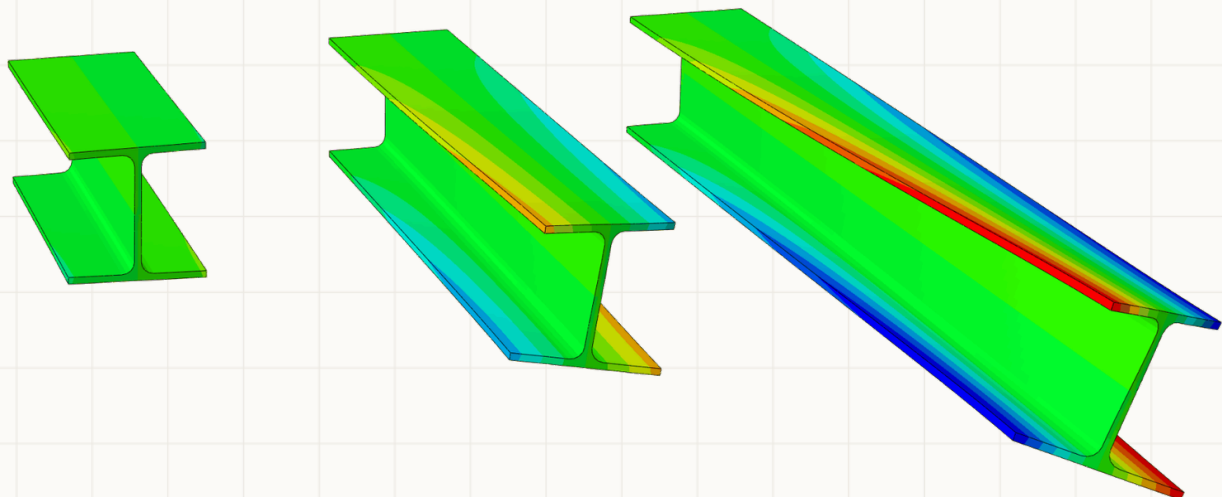


Bjørn Raeburn

Selected projects (2022-2026)

M.Sc. in Civil & Architectural Engineering (cand.polyt.), Aarhus University



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A minimal-input FE framework for advanced steel instability verification

Eurocode 3 allows an advanced, FE-based verification of steel members against instability (EN 1993-1-1 §6.3.4), but it is rarely used in practice because it demands a lot of modelling. We built a solver that makes the modelling decisions itself.

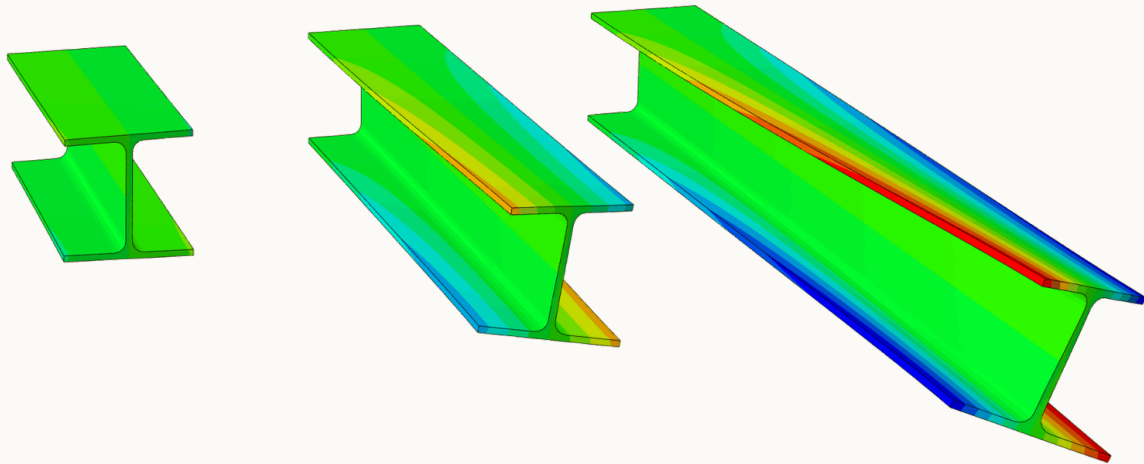
MATLAB

Abaqus (solid & 7DOF beam)

FEM-Design

EN 1993-1-1 §6.3.3 / §6.3.4

With Arben Tahiri & Siavash Gourabi • supervisor Johan Clausen • in collaboration with NordIngeniør / StructuralTools



1 Abaqus solid-element models at the onset of lateral-torsional buckling: the kind of high-fidelity model the self-developed solver was checked against before it was trusted.

WHAT I DID

Together with two fellow students I wrote a finite element solver from scratch in MATLAB, built around a 7DOF Bernoulli–Euler beam element with Vlasov warping. The seventh degree of freedom is what lets a beam element capture restrained warping and the bimoment: effects that ordinary 6DOF elements miss completely, and which can dominate the response of open thin-walled steel sections. On top of the element sit three analysis paths: a linear static solve, an eigenvalue buckling solve, and a nonlinear elastoplastic solve with an implicit return-mapping von Mises plasticity model at fibre level.

The core of the project is §6.3.4, which needs two load amplifiers, $\alpha_{ult,k}$ and $\alpha_{cr,opt}$ that the code only defines loosely. We automated their determination: a secant-slope criterion on the nonlinear load–displacement path, and a classification of the eigenmodes that picks the first out-of-plane buckling mode. The result was wrapped in an interactive MATLAB app so the engineer can see and adjust the choices instead of trusting the program blindly.

The solver was validated step by step on selected cases before it was used: displacements, rotations and section forces against Abaqus solid models under axial, bending, torsional and combined loading; buckling loads and modes against Abaqus beam models and Euler's formula, plus the elastic critical moment against the three-factor formula for the whole HEA family; and the elastoplastic response of a cantilever against Abaqus solid and beam models and a hand-calculated collapse load.

WHAT I TOOK AWAY

Steel is the material I have worked most with. Torsion, warping and lateral-torsional buckling went from being the hard chapter to something I can calculate and implement.

A design code has a lot of logic between the lines. Turning §6.3.4 into an unambiguous procedure meant taking a position on every unclear point and being able to justify it.

Validation before results. Every solver path was checked against Abaqus and closed-form solutions before the first utilisation ratio was reported.

IN NUMBERS

7DOF

beam element incl. warping

–35 %

utilisation vs. §6.3.3, hollow sections

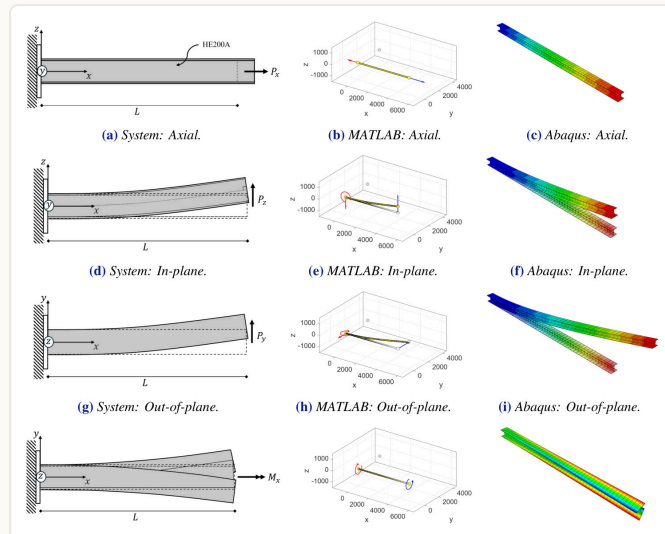
≈1.2 s

engine time per member verification

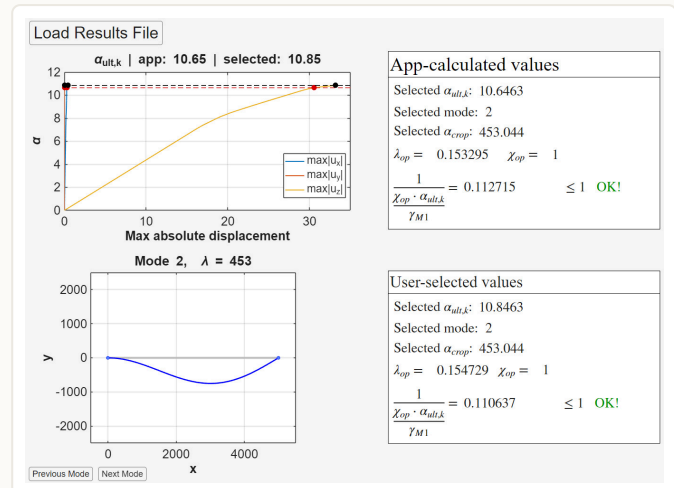
80 kg CO₂

saved on one 3 m HEB member

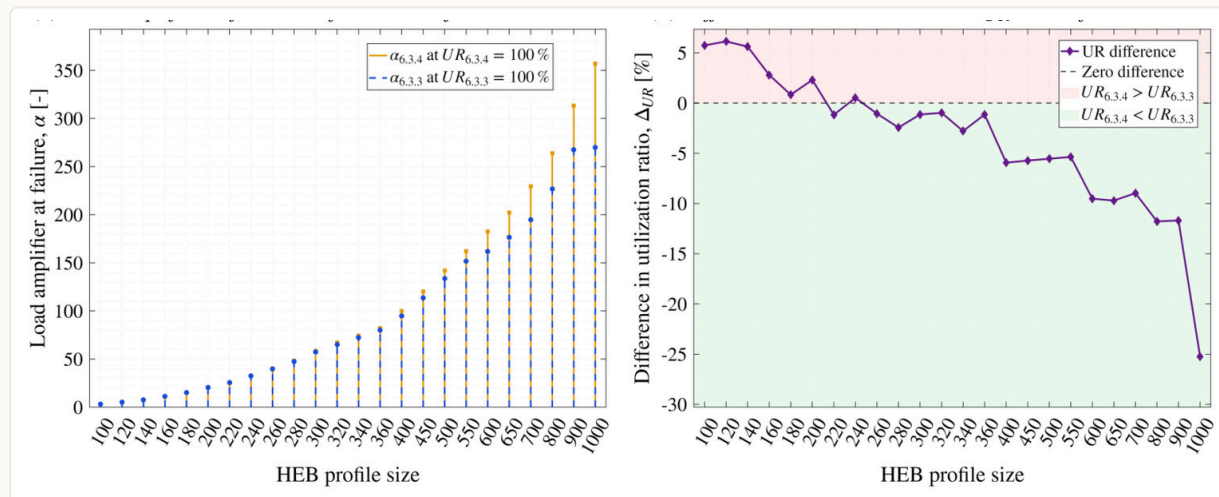
A minimal-input FE framework for advanced steel instability verification



2 Validation cases for axial, in-plane, out-of-plane and torsional loading: system sketch, the self-developed 7DOF solver, and the Abaqus solid model. Displacements agree with Abaqus and the closed-form solutions to within a few per cent.



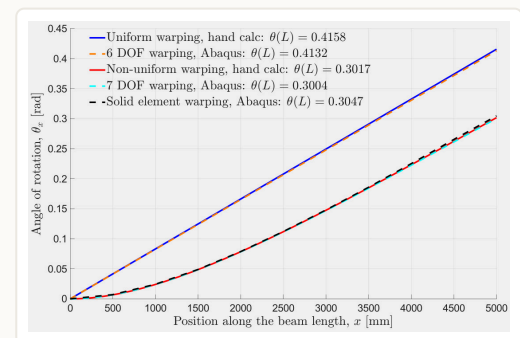
3 The interactive MATLAB app: the nonlinear α -displacement path with the automatically selected $\alpha_{ult,k}$, the buckling mode chosen for $\alpha_{cr,op}$, and the resulting §6.3.4 utilisation. App-calculated on the left, user-adjusted on the right.



4 HEB family, pinned beam: load amplifier at failure for the two routes (left) and the difference in utilisation ratio (right). The 6.3.4 verification is up to 25 % more economical for the heaviest sections.

RESULTS

To measure what the advanced method is worth, every profile in the StructuralTools database (HEA, HEB, IPE, T, RHS, SHS and channels) was loaded on a 3 m beam in combined bending and compression, once simply supported and once fixed-fixed, and the load was increased until the standard §6.3.3 interaction check and the §6.3.4 check each reported full utilisation. Compared with §6.3.3, the advanced method gave utilisation ratios up to 35 % lower for hollow sections and 28 % lower for heavy HEB members in fixed-fixed conditions, and 20–25 % lower for medium and large open profiles when simply supported. For a single 3 m HEB member that is roughly 80 kg of embodied CO₂ and 2,400 DKK of steel, with no extra modelling effort from the user: the framework takes the same input as any frame program (nodes, elements, supports, loads, section, material) and makes the remaining modelling decisions itself, including meshing, fibre layout, load eccentricity and integration order.



5 Why the seventh degree of freedom matters: a cantilevered HE200A with a torque at the free end. The 6DOF element reproduces the free-warping hand calculation, while the 7DOF element, the solid model and the restrained-warping hand calculation agree with each other and give 27 % less rotation.

Holistic: a structural engineering platform, built from scratch

Together with two of my former classmates I am building, in my spare time, the FE software we would like to use ourselves as structural engineers: analysis, Eurocode verification and calculation reports in one place. We call it Holistic because the plan is for it to cover many parts of structural design over time.

Python / NumPy

FastAPI

React / TypeScript

pytest / Vitest

EN 1990 & EN 1993 DK NA



1 The 2D frame app: bending moment diagram (drawn on the tension side, as it would be sketched by hand) and deflected shape for a 20 m portal frame, ULS combination with snow as leading action.

WHAT I DID

The first application is a 2D frame program. One can draw nodes and elements, set supports, pick materials and sections from a catalogue based on Teknisk Ståbi, apply load cases, generate load combinations to EN 1990 with the Danish annex, solve, and get a calculation report with internal forces, deflections and EN 1993-1-1 member checks with their full working. The engineering core is a pure Python package with its own linear FE solver (end releases, inclined supports, truss bars, tension- and compression-only bars). It is completely separate from the user interface, so it can be tested on its own, and every numerical function has tests against analytical solutions, textbook examples or an independent FE library.

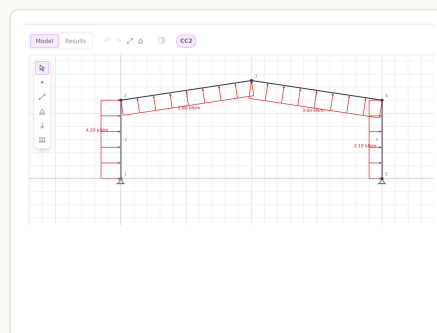
The roadmap continues with 3D frames, buckling checks and timber and concrete verification, and later the 7DOF element and §6.3.4 verification from my thesis, alongside my classmates' work on reliability-based and stochastic design.

WHAT I GET OUT OF IT

I am learning how hard it is to make a clear, usable platform for engineers: which functions matter, how they should be reached, what can be left out. That is not something you learn at university.

I get into the codes at a much deeper level when I have to automate them. Programming a clause means understanding every case it covers.

Output: what does an engineer actually want to see, and which results belong in a calculation report? Deciding that is a large part of the work.



2 Modelling the wind load case: tools down the left, input panels along the bottom, cases and combinations in their own colour.

UTILISATION		
ULS		
El.	Check	Util.
3	Bending moment y	68%
4	Bending moment y	63%
4	Combined N+M	63%
2	Bending moment y	58%
1	Bending moment y	54%
1	Combined N+M	54%
3	Shear z	14%
2	Shear z	13%

3 Utilisation summary over the drawing: the worst EN 1993-1-1 checks, worst first.

Element 4 HEB 300, 5355				
Check	Combination	Resistance	Utilisation	OK?
Tension	—	$R_{t,Ed} = A \cdot f_t / \gamma_{M0}$ EN 1993-1-1 6.2.2	—	not yet checked
Compression	ULS 1 (snow load)	$R_{c,Ed} = A \cdot f_c / \gamma_{M0}$ EN 1993-1-1 6.2.4	—	OK
Bending moment y	ULS 1 (snow load)	$M_{c,Ed} = W_{pl,y} \cdot f_y / \gamma_{M0}$ EN 1993-1-1 6.2.5	—	OK

$M_{c,Ed} = W_{pl,y} \cdot f_y / \gamma_{M0}$ EN 1993-1-1 6.2.5
 f_y 345 MPa S355 (DIN EN 10025-2)
 γ_{M0} 1.10 EN 1993-1-1 DK NA
 A 149.00 cm²
 $W_{pl,y}$ 1868.0 cm³
 $class$ 1 Table 5.2, in bending
 $M_{c,Ed}$ 371.3 kNm ULS 1 (snow load), x = 0.00 m
 $M_{c,Ed} / M_{c,Rd}$ 63 %

4 The verification in the report with the working of one check unfolded: values in, formula, resistance, utilisation.

Blast loading on a reinforced concrete element: explicit dynamics in Abaqus and LS-DYNA

The thesis idea behind this project was the 2018 Vordingborg silo demolition, where the silo fell the wrong way: could that collapse be reproduced in a simulation, and so have been predicted? The R&P project is Aarhus University's way of letting students explore a thesis topic and its software in advance, so we used it to ask the underlying question: what does it take to simulate an explosion against a reinforced concrete column, and how much does the answer depend on the concrete material model? The thesis itself ended up elsewhere, when NordIngeniør / StructuralTools proposed the steel instability topic (01) and we took it.

Abaqus/Explicit

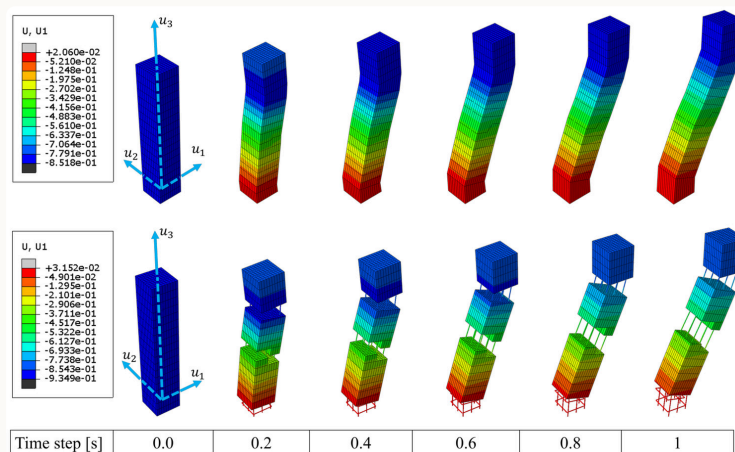
Ansys LS-DYNA

CONWEP blast model

Concrete Damage Plasticity

MAT_072R3 (Concrete Damage Rel3)

With Arben Tahiri & Siavash Gourabi · supervisor Johan Clausen



1 Reinforced concrete column hit by 8 kg TNT, calculated in Abaqus/Explicit over one second, without (top) and with (bottom) element deletion. Three failure bands develop at the base and at a third and two-thirds of the height; after that the reinforcement cage carries the load.

WHAT I DID

The blast was defined with empirical CONWEP relations (hemispherical air burst, incident and reflected pressure-time histories) and the same load was applied in both programs. In Abaqus the concrete was modelled with Concrete Damage Plasticity, calibrated from experimental compression tests, where we controlled cracking, crushing and damage evolution ourselves; in LS-DYNA with Concrete Damage Rel3, which generates most of its parameters from a minimal input. Both were run as explicit dynamic analyses with element deletion, and both models were first validated on single-element and cylinder compression tests before the column was analysed. The project also included a systematic literature review that framed the research questions.

WHAT I TOOK AWAY

I have worked thoroughly with concrete as a material through advanced material models and their failure criteria: how cracking, crushing and damage develop, and how that is described numerically in two different programs.

The two programs did not agree. Abaqus gave a tip displacement of about 0.8 m and LS-DYNA about 0.29 m for the same blast, and the difference could be traced back to the material models, not the loading.

In fast, nonlinear problems the material model and its calibration govern the answer. A minimal-input model is convenient until you need to trust the number.

IN NUMBERS

8 kg

TNT, hemispherical air burst

2

explicit solvers, same load

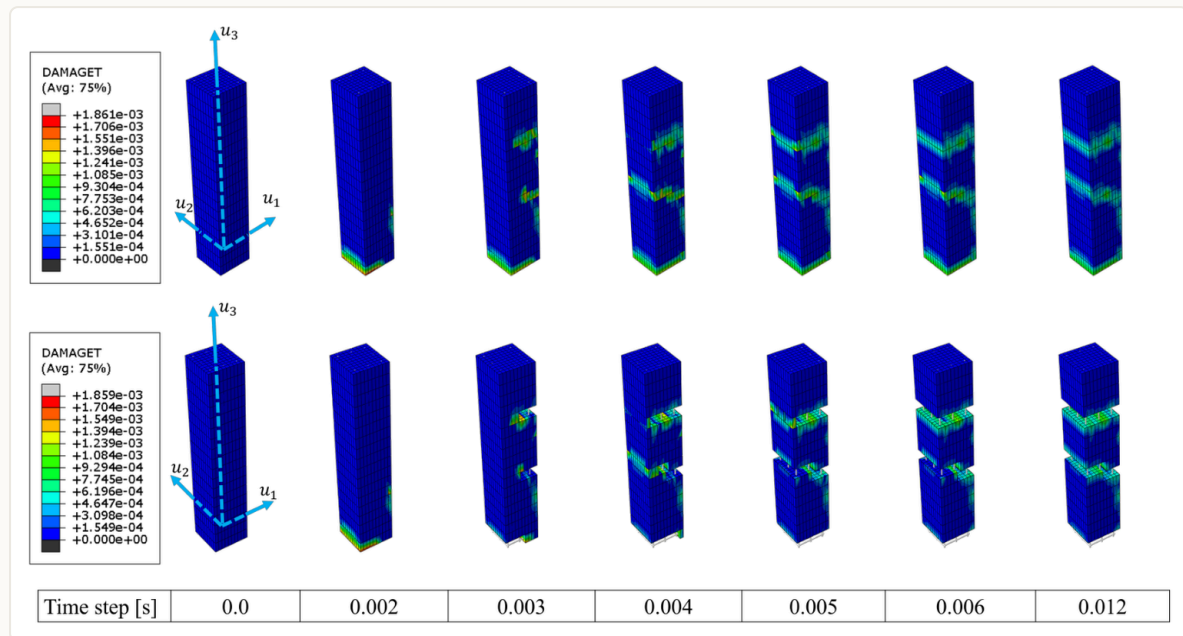
0.8 / 0.29 m

tip displacement, Abaqus vs LS-DYNA

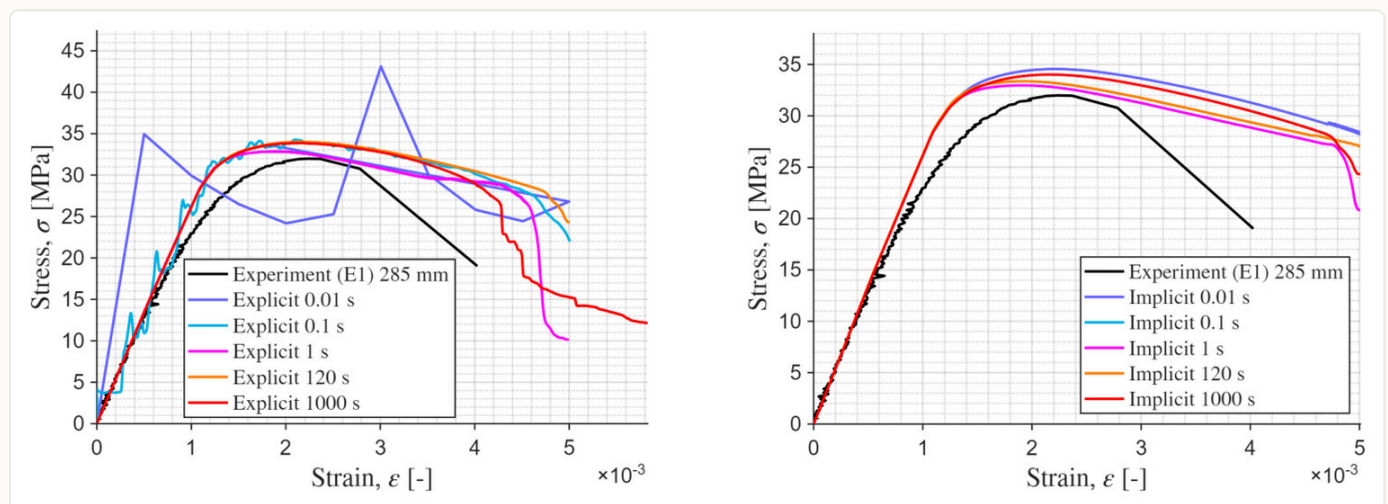
CDP / CDR3

two concrete material models compared

Blast loading on a reinforced concrete element: explicit dynamics in Abaqus and LS-DYNA



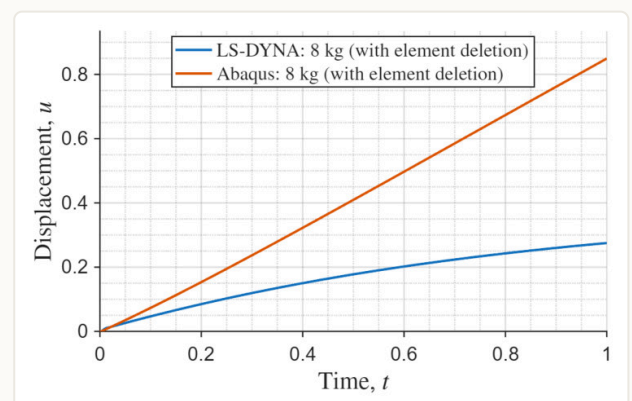
2 Tensile damage in the Abaqus CDP model during the first 12 ms: the cracks gather in the same bands that later govern the deformation.



3 Calibrating the LS-DYNA concrete model against the experimental compression curve, explicit and implicit solver, for several loading rates.

VALIDATION BEFORE THE COLUMN

Nothing was calculated on the column before it had been checked on something simpler. Both material models were run on a single element and on a concrete cylinder in compression, explicit and implicit, and compared with the experimental stress-strain curve, so we could explain the differences in the column afterwards. The Abaqus CDP model followed the test curve closely, while the LS-DYNA CDR3 model with its automatically generated parameters consistently overestimated the strength. That is exactly the difference that later gave a stiffer, less damaged column in LS-DYNA. It went straight into our recommendations for a full-scale silo model: calibrate the concrete model against test data, model fracture of the reinforcement explicitly, and plan for the computational cost of explicit dynamics from the start.



4 Tip displacement over one second: Abaqus (CDP) against LS-DYNA (CDR3) with element deletion.

A parametric multi-storey steel building, where I fell for Grasshopper

Not one design but a solution space: a three-storey steel frame whose footprint, storey offsets, column grid, joint stiffness and foundations are all parameters, and where forces, cost and CO₂ update immediately.

Rhino 7 / Grasshopper

Custom Python components

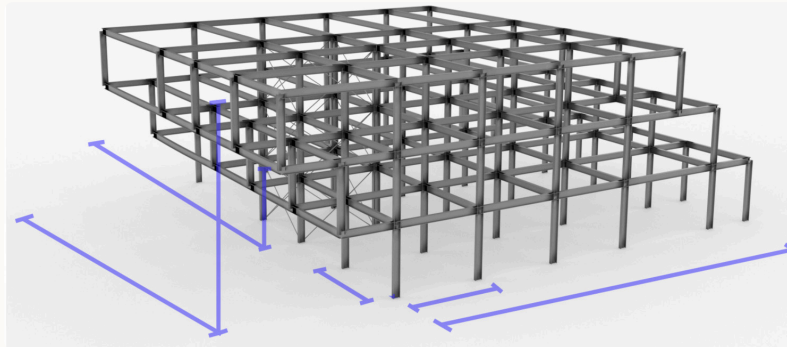
MATLAB

Optum G2

Revit

EN 1990 / 1991 / 1993

Group of six • supervisor Lars Damkilde



1 The final parametric model in Grasshopper. The blue lines mark what can be varied: footprint, storey heights, the offset of the upper floors, the grid division in both directions, and the rotational stiffness of the joints.

WHAT I DID

The building is inspired by staircase steps: each upper storey can slide up to 8 m relative to the one below. The whole structure was built and solved in Grasshopper, so footprint, storey heights, offset, column grid and joint stiffness are sliders, and internal forces, profile utilisation, cost and CO₂ update the moment one is moved. That is what made it possible to map out solution spaces instead of checking single designs. The offset made the column grid the interesting task: which ground-floor columns can continue up through an offset storey, and how do you make sure columns stand on columns instead of landing on beams? I solved it with custom Python components in Grasshopper that generate the grid points and sort the columns automatically when the parameters change.

Joint stiffness was parametrised from pinned (0) to rigid (80 000 kNm/rad) to see where the frame changes behaviour, and loads, cost and CO₂ per profile were tied to the model so the consequence of every slider could be seen. Structural design followed EN 1990, 1991 and 1993 with the Danish annexes in consequence class CC2.

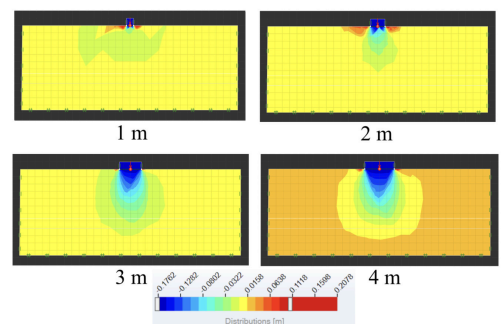
Geotechnics in the project

The foundations got their own solution space. CPT soundings from the site investigation were interpreted layer by layer into undrained shear strength and effective friction angle for the clay and sand, and the characteristic values were determined both as 5 % fractiles and as robust means (the latter was used because the data set had outliers). Strip foundations of different widths were then analysed in Optum G2, and the results interpolated in MATLAB into a surface of width, settlement and bearing capacity. From this a secant soil stiffness could be read off for a chosen foundation width and settlement and applied as spring supports under the columns in the frame model: foundation and structure in one coupled model.

WHAT I TOOK AWAY

This is the project where parametric design and programming clicked for me. Once the model is parametric, asking what if costs nothing: profile, offset, joint stiffness, cost and CO₂ are one slider away.

It was also my first proper piece of geotechnics: from raw CPT data to strength parameters, and from there to a soil stiffness the structural model could actually use.

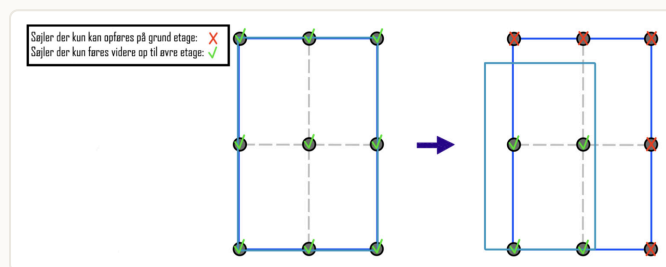


2 Optum G2: vertical deformation under strip foundations of 1, 2, 3 and 4 m width, the analyses behind the foundation solution space.

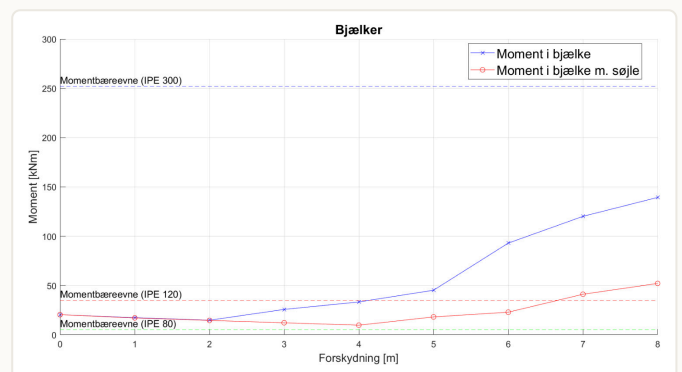
A parametric multi-storey steel building, where I fell for Grasshopper



3 The parametric study of the storey offset: start configuration with the floors aligned (left) and end configuration with the upper storeys shifted 8 m (right), with supporting columns under the overhang.



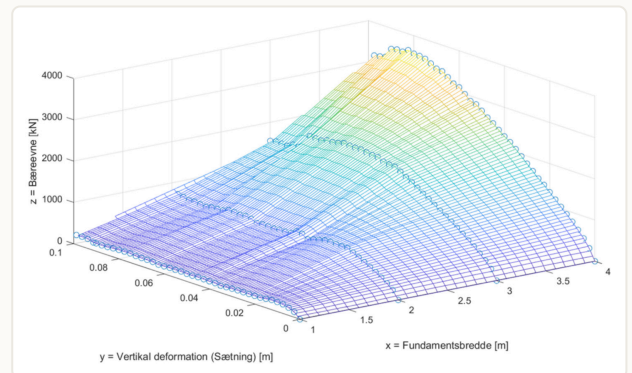
4 Sorting the columns, seen from above: which grid points can carry a column up into the offset storey (green) and which cannot (red). Handled by a Python component.



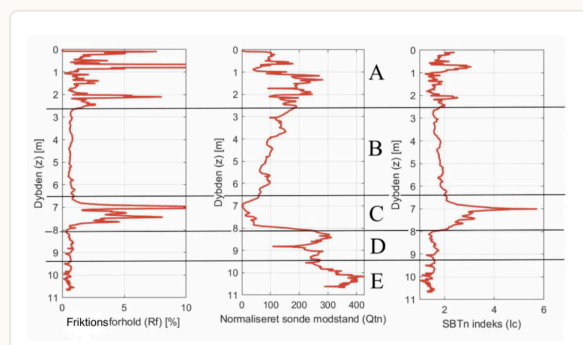
5 Critical beam moment as the upper storeys are shifted 0–8 m, with and without supporting columns under the overhang, against the capacity of IPE 80, 120 and 300.

WHAT THE SOLUTION SPACE IS FOR

The value of the model is not one answer but the overview of all the answers. Shifting the upper storeys from 0 to 8 m makes the critical beam moment roughly three times larger; supporting the overhang with columns halves it again, and in both cases IPE 120 is the smallest profile that works. That tells the architect what the staircase idea costs in steel, money and CO₂ before anyone has drawn a detail. The same goes for the ground: once the CPT data has been turned into characteristic parameters and the Optum G2 results interpolated into a surface, a foundation width can be chosen from a target settlement and the spring stiffness fed straight back into the frame model. Two disciplines that are usually handled separately became two sliders on the same screen.



6 The foundation solution space from Optum G2 and MATLAB: bearing capacity as a function of foundation width and settlement, used to read off a secant soil stiffness.



7 Interpreting the CPT sounding: friction ratio, normalised cone resistance and SBTn index as functions of depth, divided into the layers A to E that the strength parameters were determined for (clay in C, sand elsewhere).

A 70 m² timber summer house in Blokhus, from site to statics

Small in scope but covering the whole building: site analysis and tectonics, indoor climate, a timber king-post roof analysed in MATLAB, load combinations and EC5 verification, strip foundations on dune sand and Revit drawings to finish.

Timber design, EN 1995

EN 1990 / 1991 DK NA

MATLAB FEM

Geotechnics (Terzaghi)

Revit

Building physics

Group of five • supervisors Lars V. Andersen, Valentina Beatini & Julie C. Kristoffersen



1 The house as modelled in Revit: a 70 m² timber-frame summer house with exposed king-post trusses, large glazing to the south and a terrace that continues the living room outdoors.

2 The tectonic idea in a sketch: a low, open volume in the pines of Hune, activity flowing out into the landscape.

WHAT I DID

The roof was the main structural idea. We compared scissor and lattice trusses before choosing a king-post system that keeps the vapour barrier intact, set up three variants in a MATLAB FE script, and picked the best one on forces, deflection and tectonics. From there the statics were done properly: Eurocode load combinations for dead, snow and wind load, EC5 verification of rafters, purlins and wall plates for shear and for combined bending and axial force, and a deflection limit of $l/125$ that we argued for ourselves.

On the building-physics side we checked thermal comfort, air change and moisture, and it was the moisture calculation that showed a vapour barrier was needed.

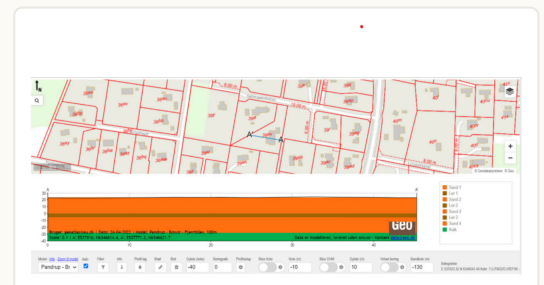
Geotechnics in the project

The site investigation showed a flat plot of aeolian sand with a thin clay layer and a groundwater table 4.7 m below ground in the nearest boring. The risk of a secondary water table in the hollows after heavy rain was one of the reasons for where the house was placed. The strip foundations under the four outer walls were checked with Terzaghi's bearing-capacity formula (sand, $\phi = 33^\circ$, $\gamma' = 9 \text{ kN/m}^3$, 1.3 m of overburden), giving a bearing capacity of 110 kN/m² against a load of 8.94 kN, and settlements were estimated from the effective stress increase and a modulus of 10 000 kPa for the sand.

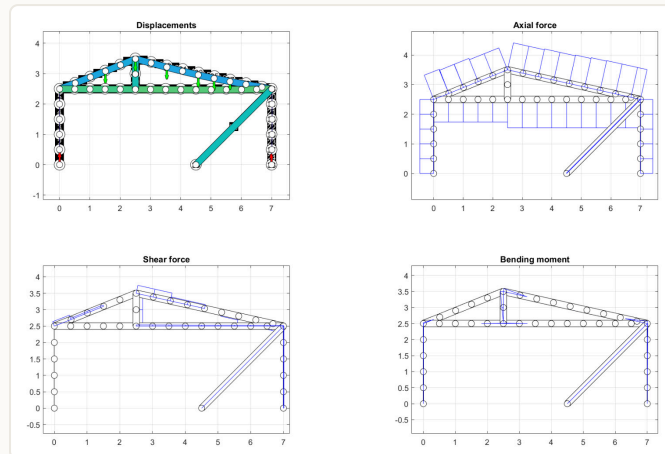
WHAT I TOOK AWAY

My favourite of the early projects because it was so concrete: a house that could actually be built, where all the disciplines had to fit together.

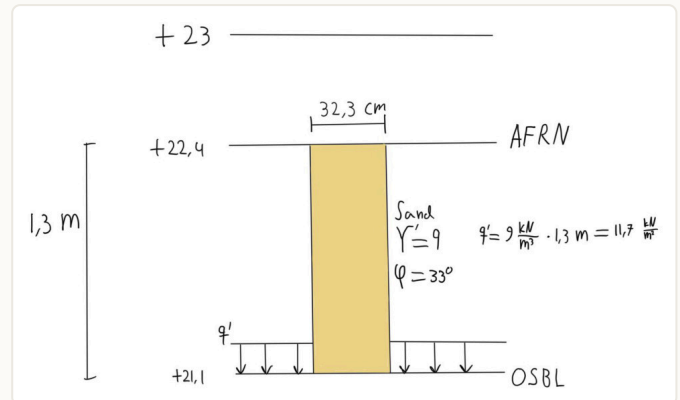
A solid grounding in timber design and practical geotechnics, and respect for how much of the quality of a small building lies in the details.



3 Terrain profile through the plot from the site investigation: sand above and below a thin clay layer, and flat ground.

A 70 m² timber summer house in Blokhús, from site to statics

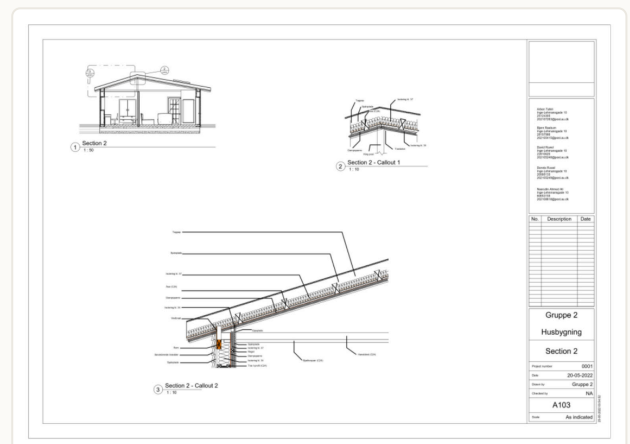
4 Internal forces from the MATLAB FE model of the king-post frame with the stabilising wall: displacements, axial force, shear and bending moment for the snow and wind cases.



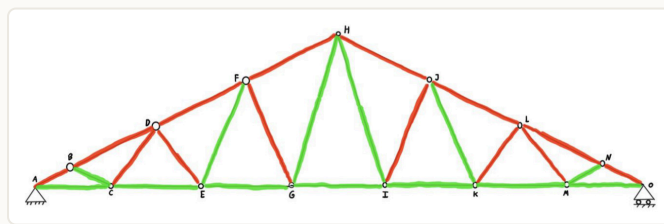
5 The strip foundation on dune sand: effective unit weight, friction angle and overburden used in Terzaghi's bearing-capacity formula.

FROM THE ROOF TO THE GROUND, AND BACK TO THE DRAWINGS

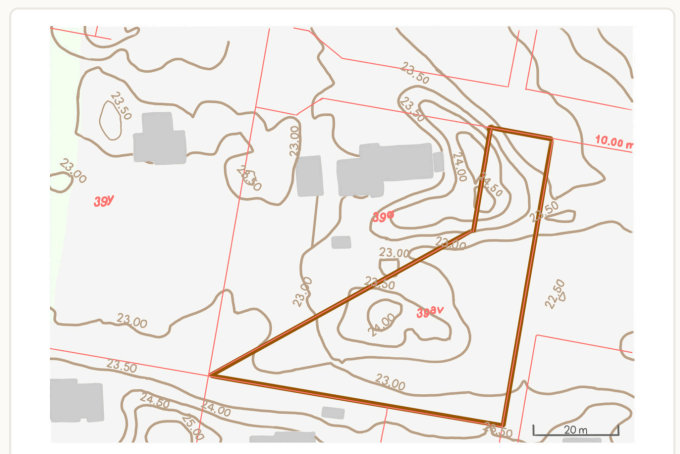
The internal-force plots from the MATLAB model settled the roof: the king-post frame carries snow and wind mostly in axial force with small bending moments, which is why slender visible timber members could be verified to EC5 with a good margin. The loads were then followed all the way down, from rafters to purlins to the strip foundation, which was checked for bearing capacity and settlement on the dune sand found in the site investigation. The Revit sheets brought it all together: plans, sections and call-out details of the places where vapour barrier, insulation and structure have to meet. It was the first time I saw a whole building hang together from concept to detail.



6 Revit sheet: section through the house with call-outs of the roof-to-wall and wall-to-ground details.



7 Tension (green) and compression (red) members in the lattice-truss alternative, one of the comparisons that led to the king-post system: fewer members, fewer penetrations of the vapour barrier and a cleaner expression.



8 Contour lines over the plot (matr. 39av): the hollows where rainwater collects were one of the reasons for where the house was placed.